

SUMMARY

A motivated student at the Next Generation Tech Booster program with experience in product design and development, CAD modeling, assembly parts, quality testing, computer vision, machine learning object detection, FE Analysis, and seeking opportunities in medical imaging, medical tech., Computer Vision, IoT solutions, Robotics, and AI.

EDUCATION

Bertelsmann Next Generation Tech Booster program
Udacity

Dec. 2024 - Present

NYU Tandon Bridge Program
New York University, New York, NY

Jan. 2024 - March 2024

B.S.E., Biomedical Engineering
Arizona State University, Tempe, AZ

May 2017

SKILLS

Software: MatLab, LabVIEW, SolidWorks, Siemens NX, ANSYS, PSpice, ImageJ, MS Word, Excel, PowerPoint, Project, Arduino IDE, and Visual Studio

Hardware: Arduino, Particle Photon, Sony Spresense, ESP32, ESP8266, and Raspberry Pi

Programming Languages: C++, C, Python, Java, and MatLab

Languages: Chinese (Native), and Spanish (Elementary Proficiency, basic, entry-level)

Others: Good verbal and written communication skills, 3D printing, assembling, troubleshooting, validation and documentation, FEA and failure analysis, technical writing

WORK HISTORY

Star Cloud Inc, New York, NY

05/2022 - 05/2023

Mobile Associate | Electronic testing technician | Sales rep. | Technical support

- Technical Expertise: Demonstrated exceptional technical proficiency by resolving customer issues related to various devices, including phones, tablets, and consumer electronics.
- Technical Troubleshooting: Successfully diagnosed and resolved complex hardware and software problems, in addition to performing essential device maintenance and repairs, ensuring optimal device functionality.
- Electronics Troubleshooting: Proficient in troubleshooting electronic circuits down to the component level, showcasing an in-depth understanding of electronics and the ability to address intricate technical issues.
- Problem Solving: Consistently diagnosed issues and proposed effective solutions, contributing to a smoother workflow and improved device performance.
- Multitasking: Demonstrated the ability to efficiently handle and prioritize multiple projects simultaneously, emphasizing strong organizational and time management skills.

IMAV LLC– Phoenix, AZ

05/2017 - 04/2020

Mechanical Design Engineer & CFD Analyst

- Collaborated with a cross-functional engineering team to design components and assemblies using SolidWorks for a gesture-controlled Unmanned Aerial Vehicle (UAV) drone chassis, focusing on aerodynamic performance and structural integrity.
- Managed and maintained Bill of Materials (BOMs) to ensure accurate inventory tracking and production efficiency.
- Prepared detailed engineering drawings with comprehensive labeling, facilitating seamless manufacturing processes and providing clear documentation for managerial and senior engineering review.
- Played a key role in drafting a proposal for the National Science Foundation (NSF) on Small Business Innovation Research (SBIR) projects, contributing to funding opportunities for innovative aerospace solutions.
- Conducted Computational Fluid Dynamics (CFD) simulations in ANSYS Fluent to analyze and optimize combustion systems, enhancing combustion efficiency and thermal performance.
- Assisted senior engineers in analyzing combustion system data to determine the full lifecycle of each design iteration, contributing to continuous improvement efforts and post-processing enhancements.
- Employed NX-10 to label and prepare design components before exporting them to ANSYS Fluent for mesh refinement, ensuring precise simulations and data accuracy.
- Demonstrated versatility by creating 3D-printed prototypes for N95 masks during the COVID-19 pandemic, showcasing a commitment to community well-being and a broad range of engineering skills.

Intelligent Embedded Systems Laboratory (IeSL), ASU, Tempe AZ

02/2016 - 03/2017

Undergraduate Research Assistant in Mechanical Engineering

- Conducted comprehensive analysis of the Hawk Moth model's lift and drag characteristics utilizing transient turbulent flow analysis, building upon prior research.
- Executed steady-state simulation analyses on a simplified 3D Hawkmoth model, comparing it to the physical Hawkmoth for performance evaluation.
- Successfully executed simulations on high-performance computing clusters to validate lift and drag coefficients across varying angles of attack.
- Designed and developed a validated 3D Hawkmoth model specifically tailored for wind tunnel testing.
- Implemented grid convergence index (GCI) case studies to ensure that lift and drag solutions achieved mesh independence, demonstrating the numerical stability of the solutions.
- Proficiently ran computational fluid dynamics (CFD) simulations on cloud computing platforms to optimize computational efficiency.
- Collaborated with a multidisciplinary team of engineers, sharing findings and insights to support the development of innovative aerospace technologies.

Arizona Biomedical Collaborative, University of Arizona College of Medicine-Phoenix, AZ

05/2016 – 08/2016

Product Design Engineer Intern

- Collaborated closely with researchers to conduct comprehensive tests on a prototype, focusing on functionality, reliability, and compatibility assessment.
- Conducted precision testing to analyze the rotational speed and direction of a visual tracking drum, contributing to data-driven design improvements.
- Created Computer-Aided Design (CAD) models for a voltage regulator while integrating various electronic components, demonstrating proficiency in design and electronics.
- Worked synergistically within a team of four, including the lab manager and researcher, to develop a visual acuity device tailored for measuring abnormal eye movements, and conducting quantitative analyses of retinal degeneration in mice.
- Demonstrated strong troubleshooting skills by addressing issues related to the motors responsible for rotating the tracking drum, contributing to the device's operational reliability and precision.

HONORS & AWARDS

- Nutanix® Hybrid Cloud Scholarship
- Intel® Edge AI Scholarship
- Google Challenge Scholarship: Android Dev

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Issued by Intel · Mar 2020

Issued by Google · Jan 2018